

Department of Computer Science and Engineering

College of Engineering Poonjar

CARDIOVASCULAR DISEASE DETECTION USING CNN

Zeroth Review

Group Members

Arunkumar K A – PJR23CS015

Gaius K Thomas – PJR23CS023

Muhammed Muhsin – PJR23CS031

Georgy Joseph – LPJR23CS047

Project Guide

Ms Jabin Mathew

Head of Department

Prof Rajesh K R

Contents

- ① Abstract
- ② Introduction
- ③ Problem Statement
- ④ Literature Survey
- ⑤ Research Gap
- ⑥ Proposed Project Idea
- ⑦ Dataset Details
- ⑧ Challenges
- ⑨ Future Scope
- ⑩ Conclusion
- ⑪ References

Abstract

- Heart sounds contain important information about the condition of the cardiovascular system.
- Traditional auscultation using a stethoscope depends heavily on the experience and listening ability of healthcare professionals, which can make the identification of abnormal heart sounds challenging.
- This project proposes an AI-Based Heart Sound Classification System that uses phonocardiogram (PCG) recordings and deep-learning techniques to automatically analyze and classify heart sounds.
- The proposed system preprocesses heart-sound recordings to reduce noise and improve signal quality.
- The processed signals are converted into suitable representations such as spectrograms.
- A Convolutional Neural Network (CNN) is then used to automatically learn important patterns from the heart-sound signals and classify them into predefined categories.
- The system aims to provide a computer-assisted screening approach for heart-sound analysis.
- It is intended to support healthcare professionals rather than replace clinical diagnosis.

Introduction

- Heart disease is one of the major health concerns worldwide. Early identification of cardiovascular abnormalities can help in timely medical evaluation and treatment.
- Heart sounds produced during the cardiac cycle contain useful information about the condition of the heart.
- Traditional heart-sound examination is generally performed using a stethoscope.
- Although auscultation is simple and inexpensive, accurate interpretation of heart sounds requires experience and training.
- Abnormal sounds can also be difficult to identify when the recordings contain background noise or weak acoustic signals.
- Recent developments in Artificial Intelligence and Deep Learning have created new possibilities for automatic heart-sound analysis.
- Digital stethoscopes can record heart sounds as digital signals, allowing them to be processed and analyzed using machine-learning algorithms.
- The base paper demonstrated that deep-learning models such as VGG-16, 1D CNN and ResNet-18 can analyze heart-sound signals for cardiovascular disease screening.

Problem Statement

- Manual identification and classification of abnormal heart sounds is challenging because it depends heavily on the experience and listening ability of healthcare professionals.
- Heart sound recordings can also contain background noise, weak signals and variations caused by recording conditions.
- These factors can make accurate classification difficult.
- Therefore, there is a need for a computer-assisted system that can automatically process heart-sound recordings, extract meaningful patterns and classify them using deep-learning techniques.

Core Objective

To design and develop an AI-based system that can automatically analyze heart-sound recordings and classify normal and abnormal heart-sound patterns accurately and efficiently using deep-learning techniques.

Literature Survey – Part 1

Study / Approach	Input	Method	Contribution
Deep CNN based cardiac analysis	Biomedical signals	Deep CNN	Automatic cardiovascular classification
VGG-16 based approach	Heart-sound / signal representation	VGG-16	Deep feature extraction and classification
1D CNN approach	Raw biomedical signals	1D CNN	Automatic signal feature learning
ResNet-18 approach	Signal / image representation	ResNet-18	Deep residual feature learning

Observation

Deep-learning methods can automatically learn useful patterns from biomedical signals and can be applied to cardiovascular screening and heart-sound analysis.

Literature Survey – Part 2

Study	Input	Method	Research Gap / Remarks
PCG based classification	Heart sounds	CNN	Normal / abnormal heart-sound classification
Multi-lead ECG classification	ECG signals	Deep CNN	Strong cardiovascular signal analysis approach
Audio representation based approach	Heart-sound recordings	Spectrogram + CNN	Useful for automated acoustic pattern recognition
Traditional machine learning	Extracted signal features	SVM	Can provide a baseline for comparison

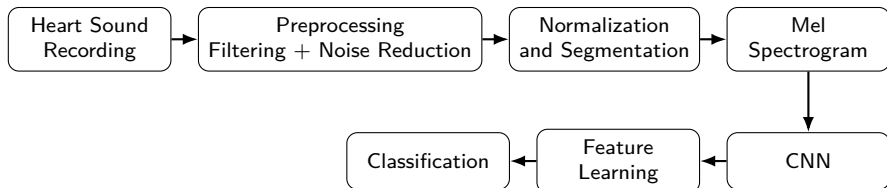
Inference

The literature indicates that both conventional machine-learning methods and deep-learning approaches can be used for biomedical signal classification. CNNs are particularly suitable for learning patterns from structured signal representations such as spectrograms.

Research Gap

- **Limited Heart-Sound Focus:** Many cardiovascular AI studies focus primarily on ECG signals, while heart-sound-based analysis receives comparatively less attention.
- **Manual Interpretation:** Traditional auscultation depends heavily on professional expertise and may be difficult when recordings are noisy or weak.
- **Noise and Variability:** Heart-sound recordings can vary because of recording devices, environments and signal quality.
- **Limited Dataset Size:** Small datasets can make deep-learning models vulnerable to overfitting.
- **Generalization:** Models trained using controlled datasets require validation on diverse real-world patient recordings.
- **Clinical Validation:** A research prototype cannot directly establish clinical diagnostic accuracy without extensive patient-based validation.

Proposed Project Idea



Proposed Approach

Heart-sound / PCG recordings are preprocessed and transformed into a suitable time-frequency representation. A CNN is then trained to automatically learn acoustic patterns and classify the recordings into predefined cardiac sound categories.

Dataset Details

Primary Dataset

HLS-CMDS – Heart and Lung Sounds Dataset Recorded from a Clinical Manikin Using Digital Stethoscope

- Contains approximately **535 recordings**.
- Includes heart sounds, lung sounds and mixed cardiopulmonary recordings.
- Recordings are available as WAV audio files with associated metadata.
- The proposed project primarily focuses on the **heart-sound recordings**.
- The recordings were obtained using a clinical manikin in a controlled environment.
- The dataset provides a useful starting point for heart-sound classification research.
- The controlled nature of the dataset should be considered when evaluating model generalization.

Important Limitation

Manikin-based results alone cannot confirm real-world clinical accuracy.

Challenges

- **Limited Dataset**

Difficulty obtaining large and diverse labelled heart-sound datasets.

- **Noise in Recordings**

Background noise can affect prediction accuracy.

- **Data Variability**

Heart sounds differ between patients and recording devices.

- **Class Imbalance**

Some disease categories may contain fewer samples.

- **Model Overfitting**

CNN may perform well on training data but poorly on new data.

- **Computational Requirements**

Deep-learning training may require high processing power.

- **Clinical Validation**

Models require extensive testing before real-world medical use.

- **Interpretability**

Understanding why the model makes a particular prediction.

Future Scope

- **Real-Time Detection**

Enable heart-sound analysis using a digital stethoscope.

- **Mobile Application**

Develop a mobile application for easy heart-sound analysis.

- **Larger Datasets**

Train the model using larger and more diverse clinical datasets.

- **Multi-Disease Detection**

Extend the system to multiple cardiovascular conditions.

- **Cloud Integration**

Enable remote analysis and storage of recordings.

- **Explainable AI**

Provide understandable reasons behind model predictions.

- **Clinical Integration**

Integrate the system with healthcare systems after validation.

- **Continuous Monitoring**

Support long-term monitoring of heart health.

Conclusion

- The proposed AI-Based Heart Sound Classification System demonstrates the application of Artificial Intelligence and Deep Learning for automated heart-sound analysis.
- By processing PCG recordings and using CNN-based classification, the system can learn important acoustic patterns and classify heart-sound recordings automatically.
- The base paper demonstrates the feasibility of using deep-learning models such as VGG-16, ResNet-18 and CNN for analyzing cardiovascular signals.
- Our project builds on this concept by developing a practical pipeline involving preprocessing, spectrogram generation, CNN-based feature learning and classification.
- Although the proposed system is not intended to replace professional medical diagnosis, it can serve as a computer-assisted screening and research tool.
- With larger datasets, improved preprocessing, real-time digital-stethoscope integration and explainable AI, the system can be further developed into a more robust cardiovascular monitoring and screening platform.

References

- ❶ *Classification of Myocardial Infarction with Multi-lead ECG Signals and Deep CNN.*
- ❷ *HLS-CMDS: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin Using Digital Stethoscope.*
- ❸ Research literature on deep-learning-based cardiovascular signal classification.
- ❹ Research literature on phonocardiogram (PCG) signal processing and heart-sound classification.
- ❺ Research literature on CNN-based audio and spectrogram classification.
- ❻ Research literature on digital stethoscope based heart-sound analysis.

Thank You

Questions & Discussion

Project Guide: Jabin Mathew

Department of Computer Science and Engineering

College of Engineering Poonjar